

316-221 Metallurgy

CHAPTER 8: HEAT TREATMENT



Heat treatment

Heat Treatment – defined as the controlled heating and cooling of metals for the primary purpose of altering their properties (**strength, ductility, hardness, toughness, machinability, etc**)

- ❑ Can be done for **strengthening purposes** (converting structure to martensite)
- ❑ Can be done for **softening and conditioning purposes** (annealing, tempering, etc.)

Heat treatment

Steels can be heat treated to produce a great variety of microstructures and properties. Generally, heat treatment uses phase transformation during heating and cooling to change a microstructure in a solid state.

In heat treatment, the processing is most often entirely thermal and modifies only structure.



Steel castings after undergoing 12-hour 1,200 °C heat treatment.

Objectives of heat treatment

The major objectives are:

- To increase strength, hardness and wear resistance (**bulk hardening, surface hardening**)
- To increase ductility and softness (**tempering, recrystallization annealing**)
- To increase toughness (**tempering, recrystallization annealing**)
- To obtain fine grain size (**recrystallization annealing, full annealing, normalizing**)
- To remove internal stresses induced by differential deformation by cold working, non-uniform cooling from high temperature during casting and welding (**stress relief annealing**)

Objectives of heat treatment

- To improve machineability (**full annealing and normalizing**)
- To improve cutting properties of tool steels (**hardening and tempering**)
- To improve surface properties (**surface hardening, corrosion resistance-stabilizing treatment and high temperature resistance-precipitation hardening, surface treatment**)
- To improve electrical properties (**recrystallization, tempering, age hardening**)
- To improve magnetic properties (**hardening, phase transformation**)

Heat treatment of steel

Classification of steels (EN 10020)

□ Carbon steels □ Alloy steels

Inclusions, %	Carbon steels	Alloy steels
Mn	≤ 1.65	> 1.8
Si	≤ 0.5	> 0.5
Cr	≤ 0.3	> 0.5
Ni	≤ 0.3	> 0.5
Ti	≤ 0.05	> 0.12
V	≤ 0.1	> 0.12

Heat treatment of steel

<u>Carbon steels</u>	<u>Alloy steels</u>
- Low C-steels (<0.25%)	- Low alloy steels (<2.5%)
- Medium C-steels (0.3-0.6%)	- Medium alloy steels (<5%)
- High C-steels (>0.6%)	- High alloy steels (>5%)

Cast irons



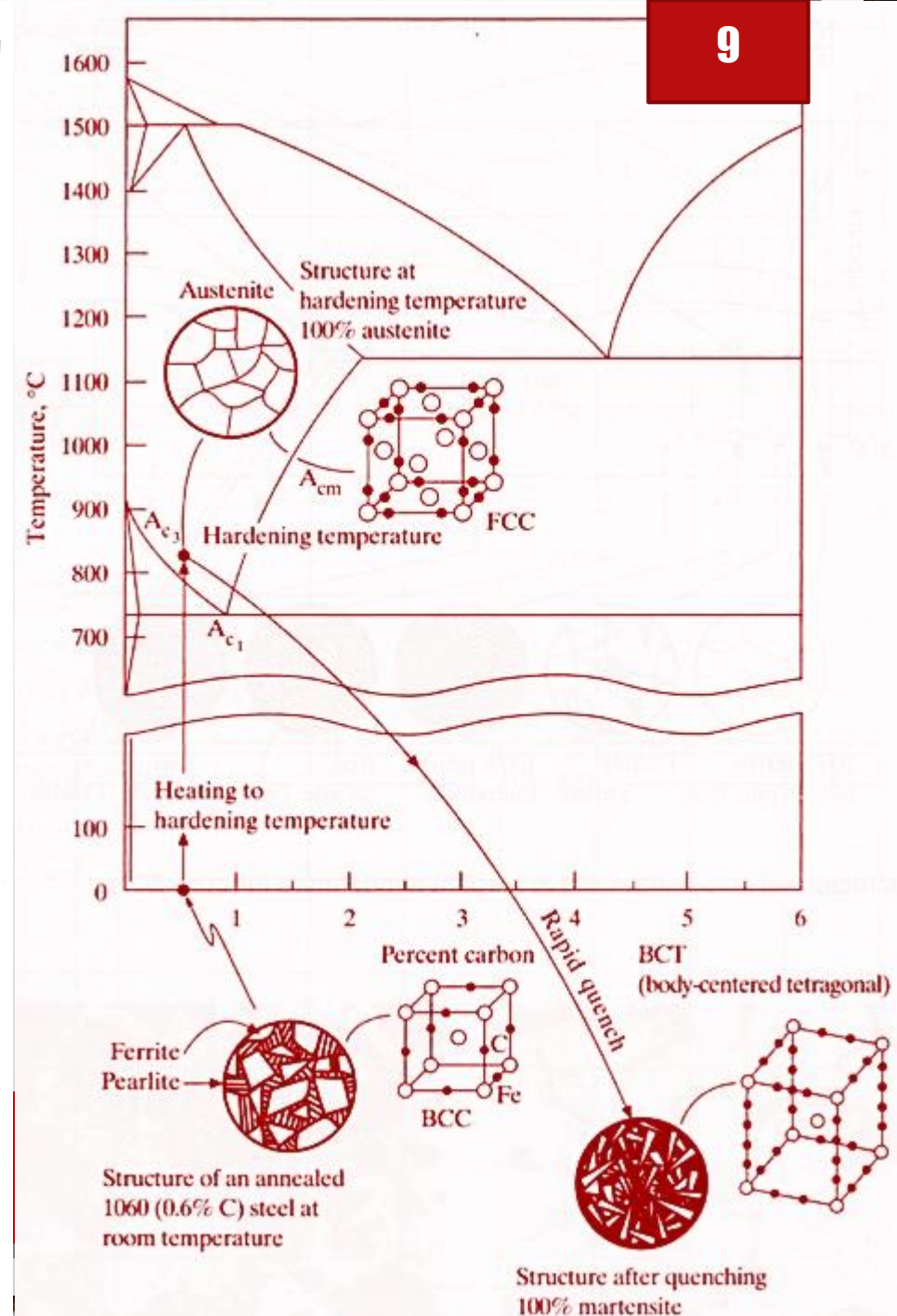
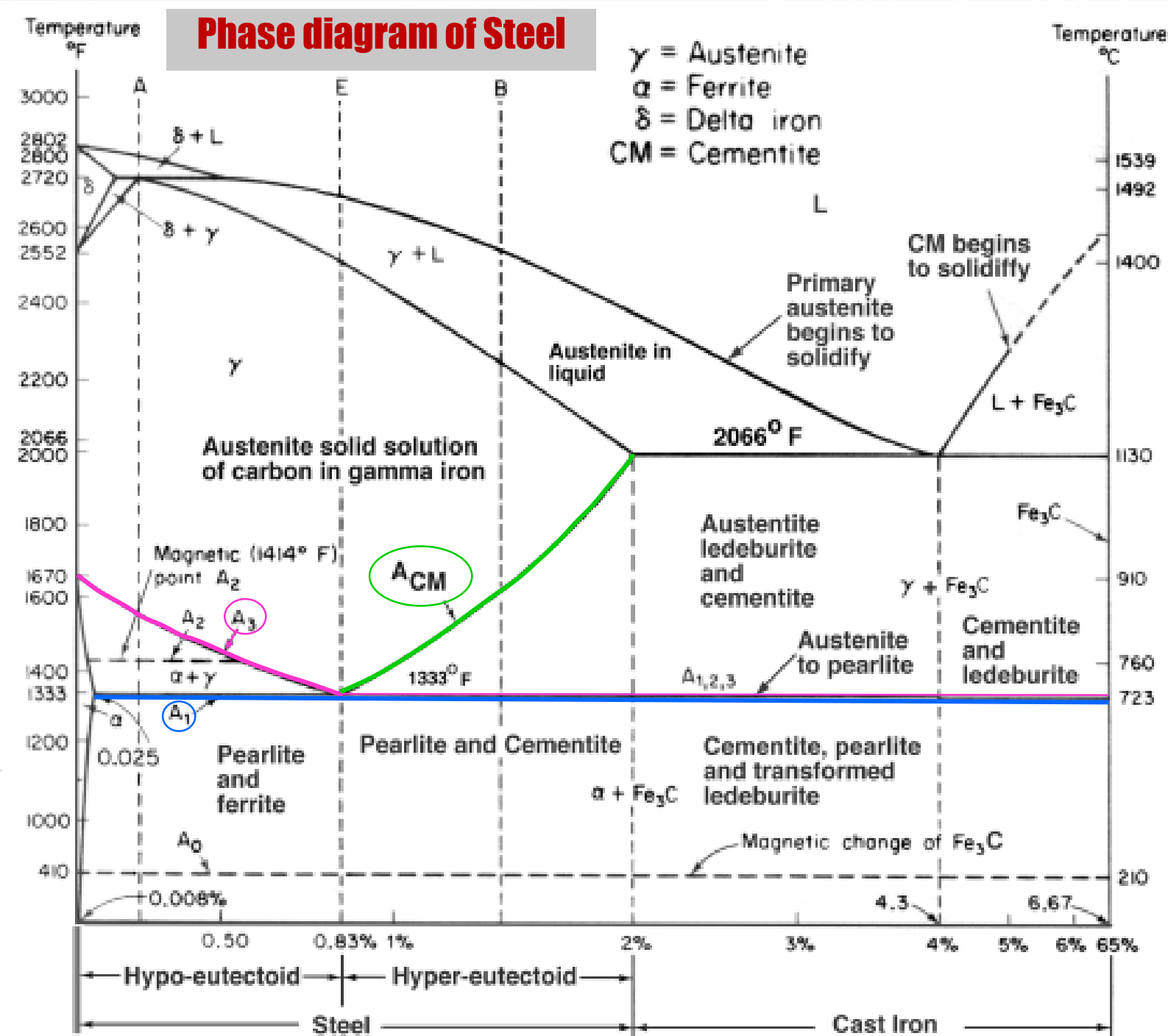
<u>Alloy steels</u>
- Cr steels
- Mn steels
- Cr-Ni steels etc.

Heat treatment of steel

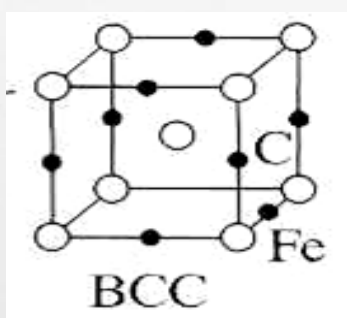
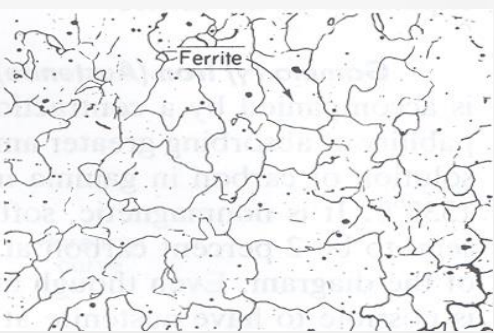
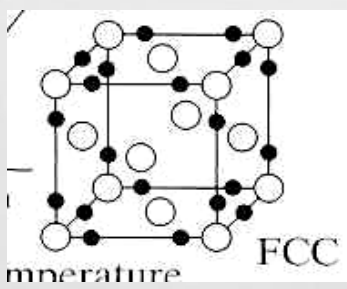
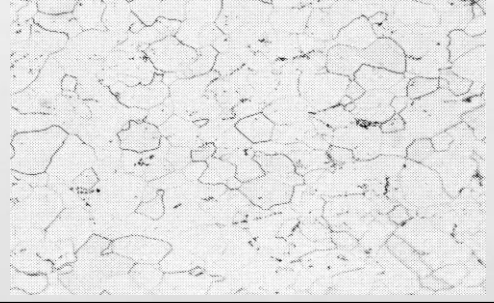
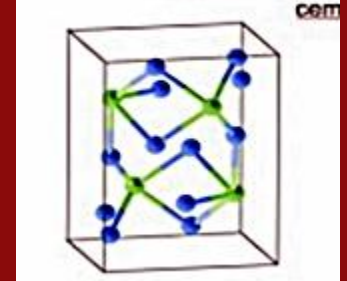
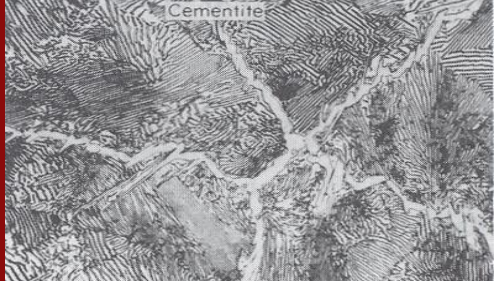
Carbon steels

- Steel = 0.06-2.0% carbon
- Must have a carbon content of at least 0.6% (ideally) to heat treat.
- Must heat to austenitic temperature range.
- Must rapid quench to prevent formation of equilibrium products.
- Basically crystal structure changes from **BCC** to **FCC** at high Temp.
- The FCC can hold more carbon in solution and on rapid cooling the crystal structure wants to return to its BCC structure. It cannot due to trapped carbon atoms. The net result is a distorted crystal structure called body centered tetragonal called **martensite**.

Phase diagram of Steel

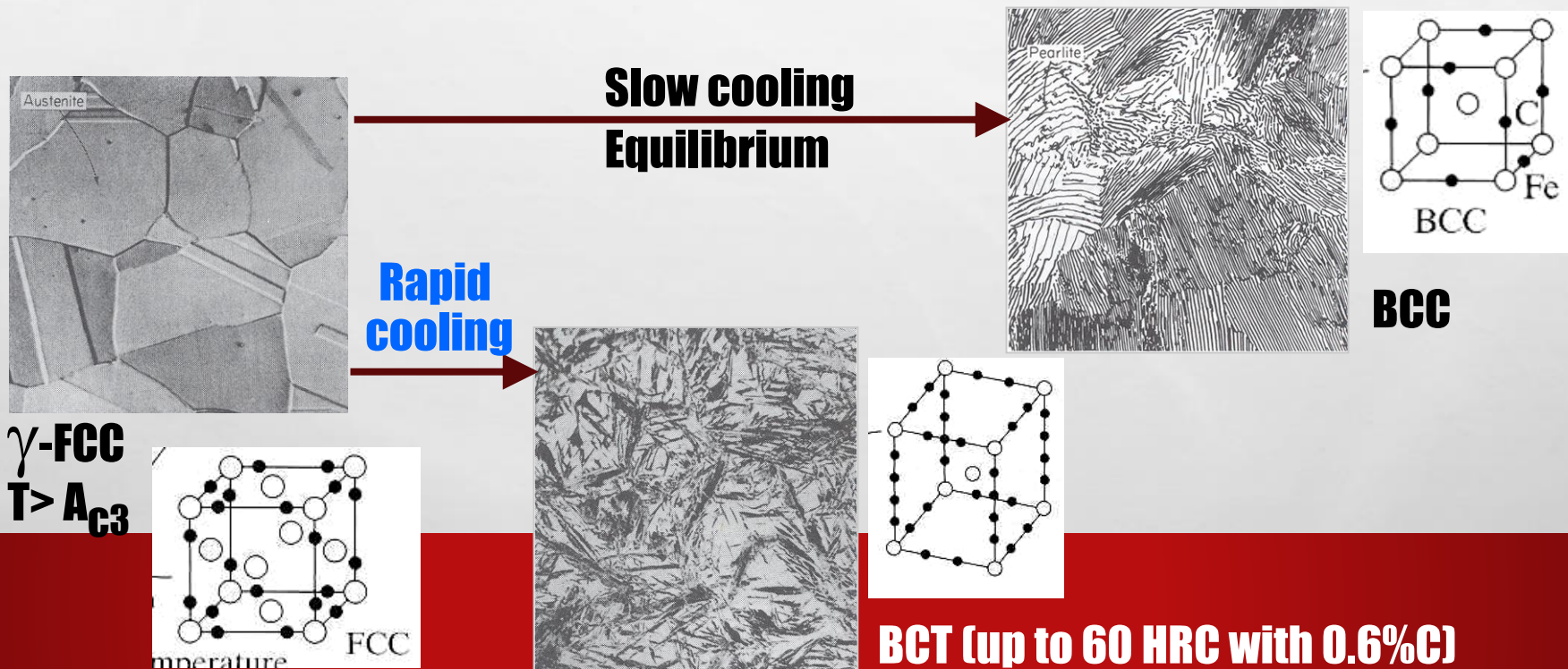


Morphology of Steel

Phase	Crystal structure		Micrograph	Characteristic
Ferrite	BCC with carbon in solid solution			Soft, Ductile, Magnetic
Austenite	FCC with carbon in solid solution			Soft, Moderate strength, nonmagnetic
Cementite	Compound of iron and carbon (Fe_3C)			Hard and Brittle

What happens when a steel is heated or cooled under nonequilibrium condition?

- On heating, phase change as in Phase Diagram.
- On rapid Cooling, C atom in γ -FCC are trapped, resulting in a (Body centered tetragonal) structure which is called "Martensite"



Types of heat treatment of steel

1. Hardening
2. Tempering
3. Annealing
4. Normalizing



<http://olyk.fullmediadev.com/kiln-varieties/glass-firing-kilns/annealing-crucible-kilns.html>

Types of heat treatment of steel

1. Hardening

- ❑ Hardening steel causes the structure of carbon to crystallize, similar to the way coal or graphite changes to diamond under the heat and pressure within the earth.
- ❑ Steel to be hardened needs to have a carbon content of about 0.6% or higher. This kind of steel is called **high-carbon steel or tool steel**.

Types of heat treatment of steel

- ❑ Medium carbon steels, with a carbon content of 0.4 to 0.55 %, can also be hardened, but these steels need to have other metals alloyed with them so that the steel can be hardened more thoroughly.
- ❑ Mild steels, with a carbon content of 0.4% or less, cannot be hardened directly. They can, however, be coated with other materials to make them harder in a process called case hardening.

Types of heat treatment of steel

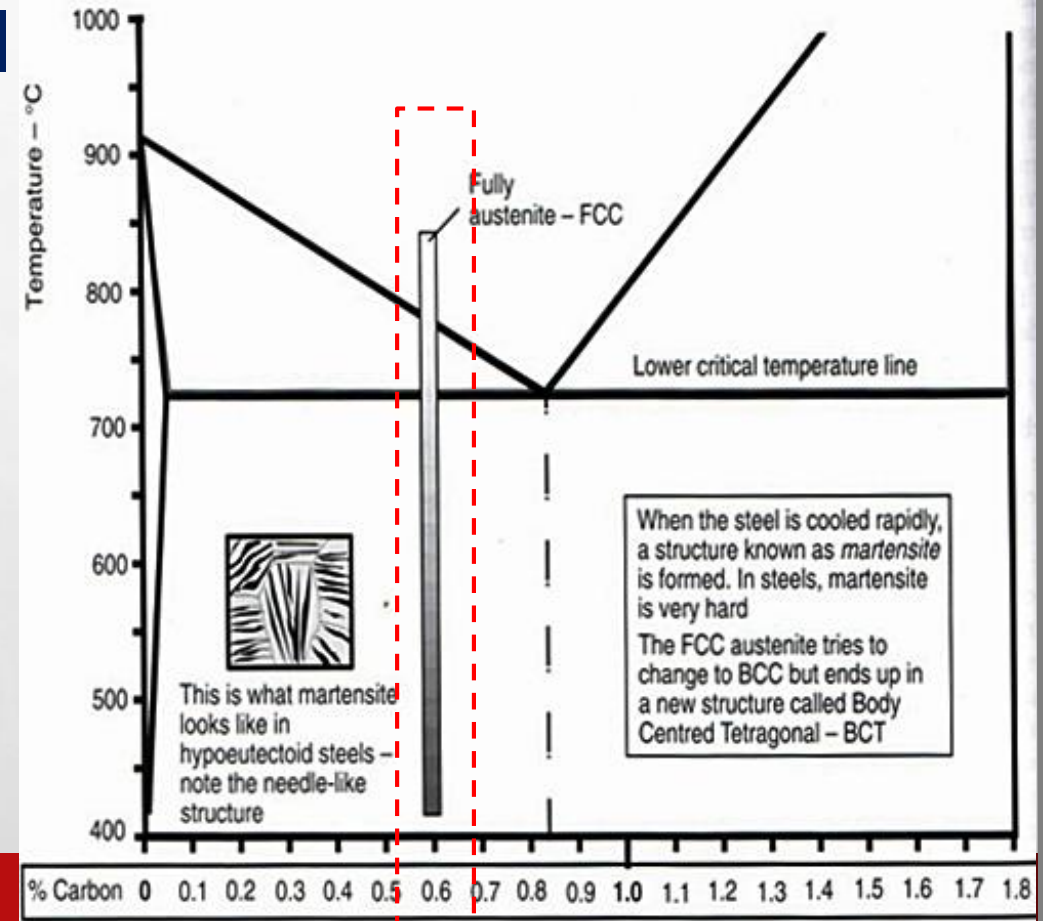
1. Hardening

Medium and high carbon steels (0.4 – 1.2%C) can be heated until red hot and then quenched in water producing a very hard and brittle metal. At 723 °C, the BCC ferrite changes into Austenite with a FCC structure.

Types of heat treatment of steel

Ex. Hardening 0.6% carbon steel

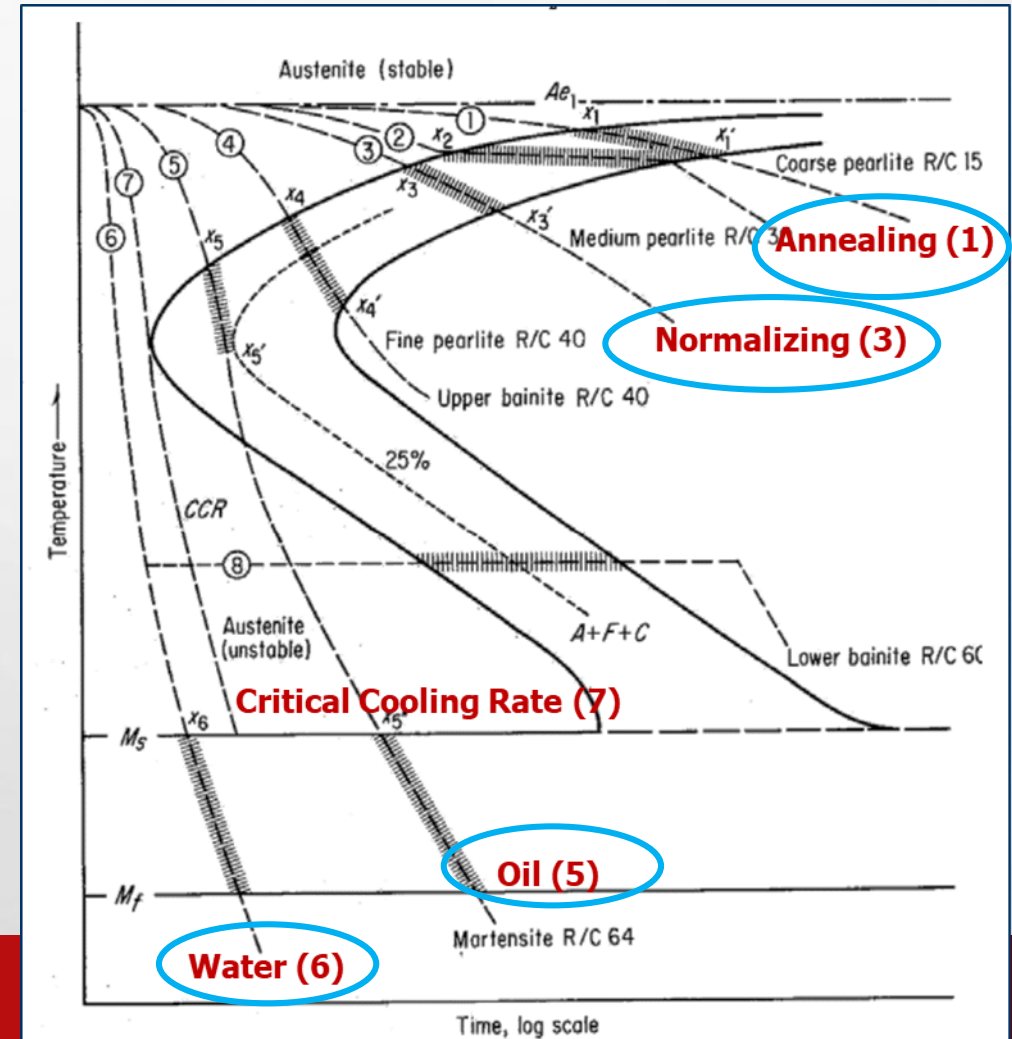
- The metal is heated to over 780 °C , which allows the carbon to dissolve into the FCC Austenite.
- Quenching the metal quickly in water prevents the structure from changing back into BCC.
- A different structure, Body Centre Tectragonal (BCT) is formed. It is called Martensite and is extremely hard and brittle with a needle-like microstructure.



Types of heat treatment of steel

Quenching media

- ☐ Brine (water and salt solution)
- ☐ Water
- ☐ Oil
- ☐ Air
- ☐ Turn off furnace



Types of heat treatment of steel

Quenching media

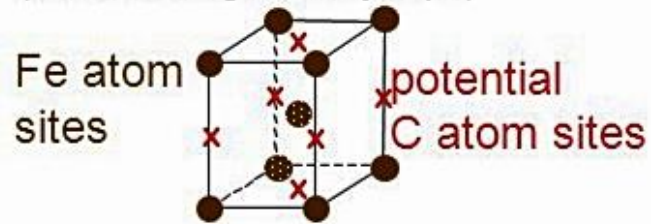
- ❑ Soaking temperature 30-50°C above A_3 or A_1 , then fast cooling (in water or oil) with cooling rate exceeding a critical value. The critical cooling rate is required to obtain non-equilibrium structure called **martensite**. During fast cooling austenite cannot transform to ferrite and pearlite by atomic diffusion.
- ❑ Martensite is supersaturated solid solution of carbon in α -iron (greatly supersaturated ferrite) with tetragonal body centered structure. Martensite is very hard and brittle. Martensite has a **“needle-like”** structure.

Types of heat treatment of steel

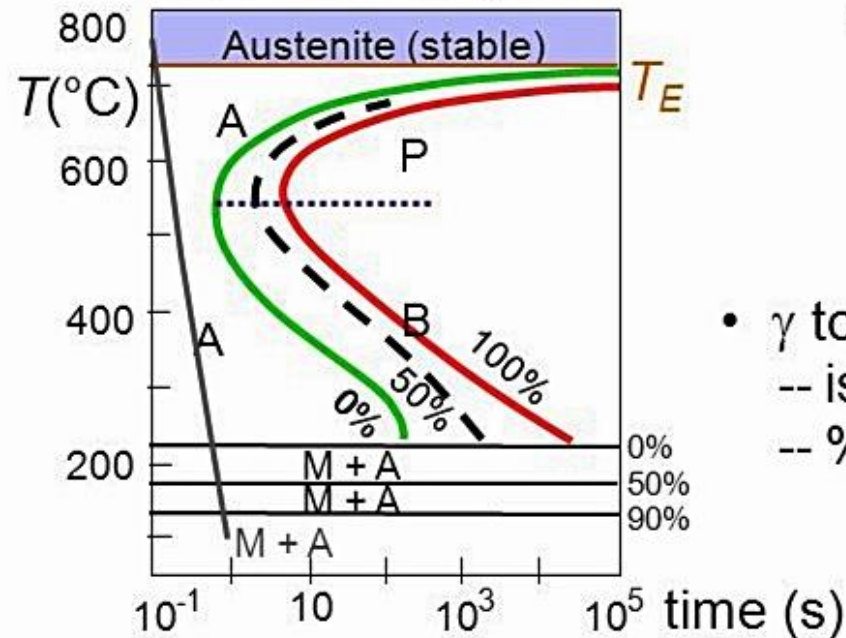
19

■ Martensite:

-- γ (FCC) to Martensite (BCT)
(involves single atom jumps)



■ Isothermal Transf. Diagram



— Martensite needles
— Austenite

courtesy United States Steel Corporation.

- γ to M transformation..
-- is rapid!
-- % transf. depends on T only.

Types of heat treatment of steel

Case hardening or surface hardening

- ❑ Low carbon steels cannot be hardened by heating due to the small amounts of carbon present.
- ❑ Case hardening seeks to give a hard outer skin over a softer core on the metal.
- ❑ The addition of carbon to the outer skin is known as **carburizing**.

Types of heat treatment of steel

Carburizing

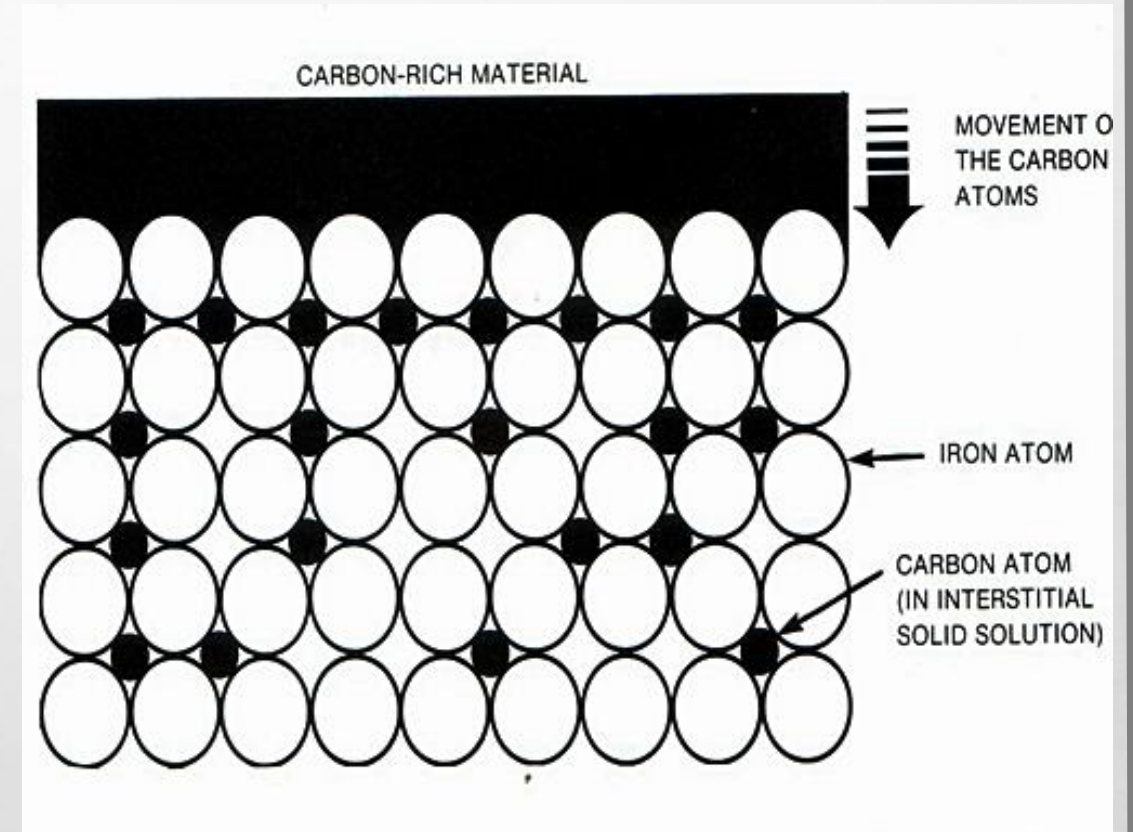
- ❑ Carburizing is a process used to case-harden steel with a carbon content between 0.1 and 0.3 wt% C. In this process steel is introduced to a carbon rich environment and elevated temperatures for a certain amount of time, and then quenched so that the carbon is locked in the structure.
- ❑ The carbon can come from a solid, liquid or gaseous source; if it comes from a solid source the process is called **pack carburizing**. Packing low carbon steel parts with a carbonaceous material and heating for some time diffuses carbon into the outer layers. A heating period of a few hours might form a high-carbon layer about one millimeter thick.
- ❑ Liquid carburizing involves placing parts in a bath of a molten carbon-containing material, often a metal cyanide; **gas carburizing** involves placing the parts in a furnace maintained with a methane-rich interior.



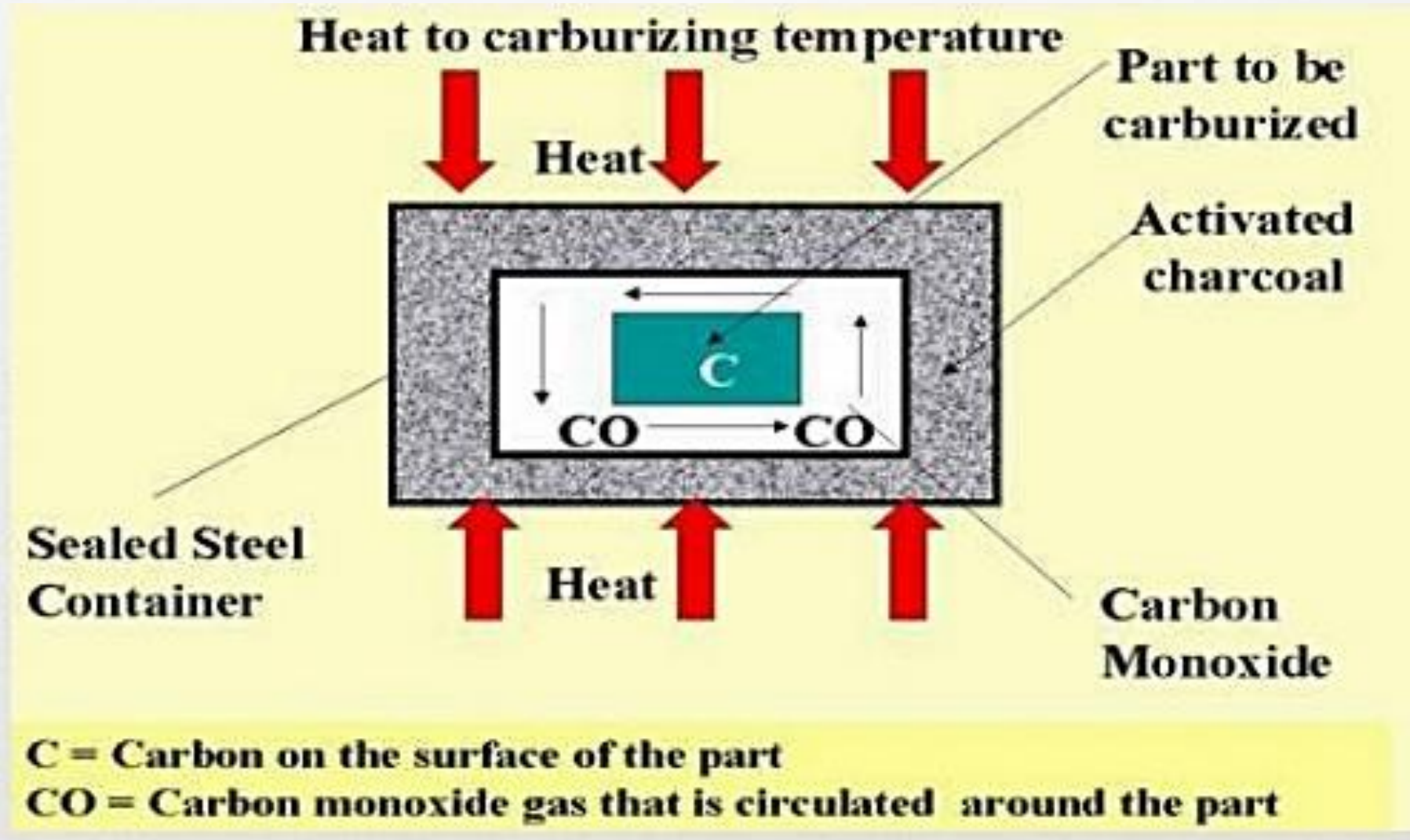
Types of heat treatment of steel

Pack carburizing

- ❑ The component is packed surrounded by a carbon-rich compound and placed in the furnace at 900 degrees.
- ❑ Over a period of time carbon will diffuse into the surface of the metal.
- ❑ The longer left in the furnace, the greater the depth of hard carbon skin. Grain refining is necessary in order to prevent cracking.



Types of heat treatment of steel



Types of heat treatment of steel

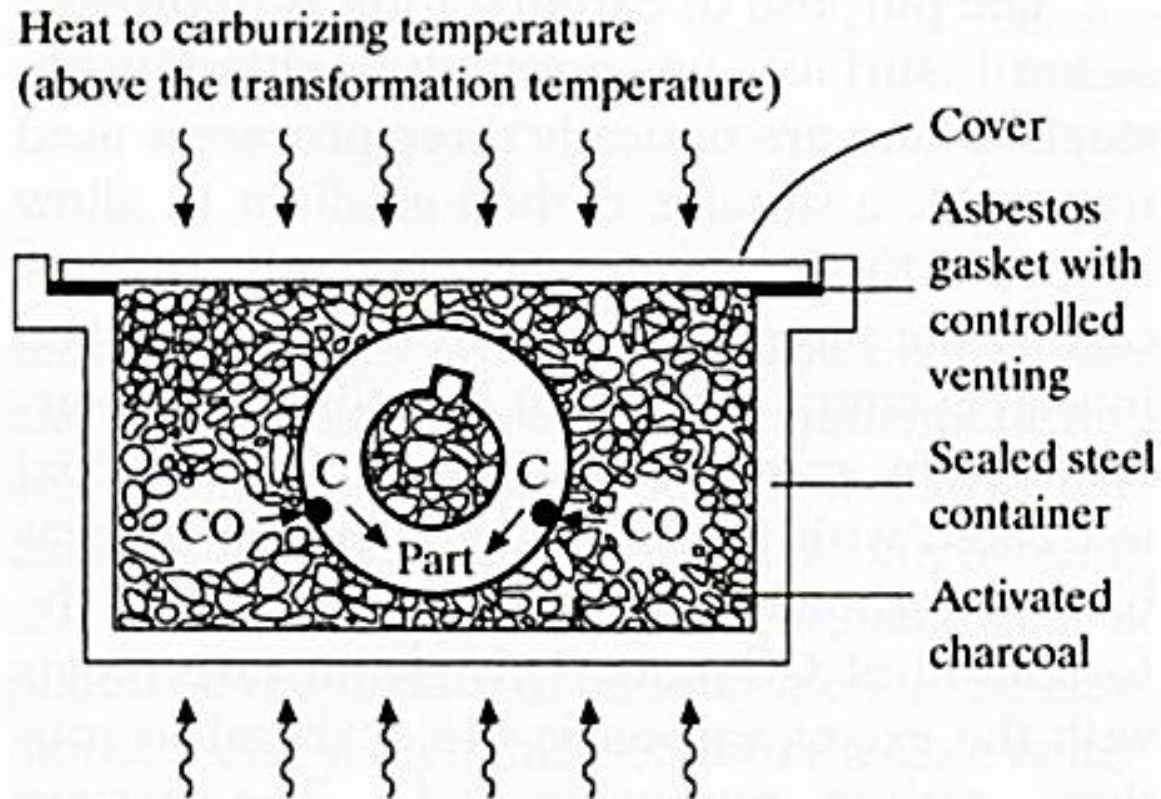


Figure 10-21
Pack carburizing

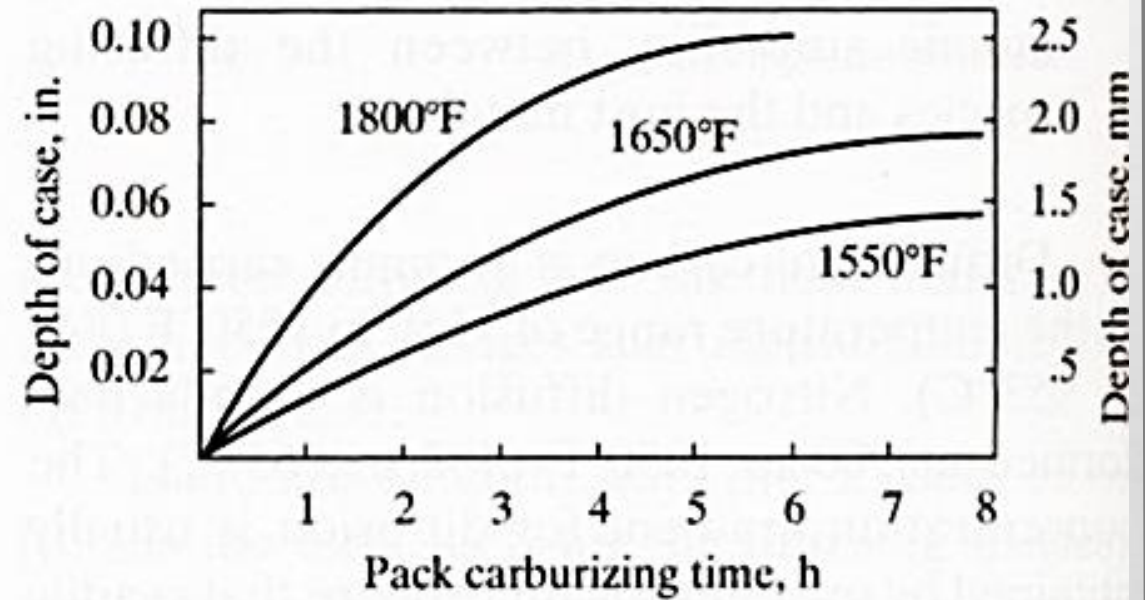


Figure 10-22
Effect of carburizing temperature on case depth
Source: G. M. Enos and W. E. Fontaine. *Elements of Heat Treatment*. New York: John Wiley & Sons, Inc., 1963.

Types of heat treatment of steel

- ❑ **Salt bath carburizing.** A molten salt bath (sodium cyanide, sodium carbonate and sodium chloride) has the object immersed at 900 °C for an hour giving a thin carbon case when quenched.

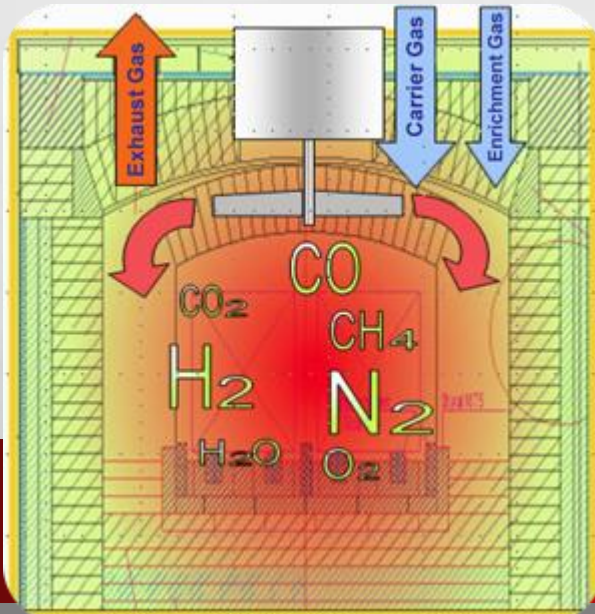


Ref. <http://www.tecmasrl.eu/en/2013/01/salt-baths/>

Types of heat treatment of steel

- ❑ **Gas carburizing** process is a surface chemistry process, which improves the case depth hardness of a component by diffusing carbon into the surface layer to improve wear and fatigue resistance.

The work pieces are pre-heated and then held for a period of time at an elevated temperature in the **austenitic region** of the specific alloy, typically between 820-940°C.



During the thermal cycle the components are subject to an enriched carbon atmosphere such carbon can diffuse into the surface layers of the component.

The rate of diffusion is dependent on the alloy, carbon potential of the atmosphere and time.



Uniform surface carburizing of a gear wheel

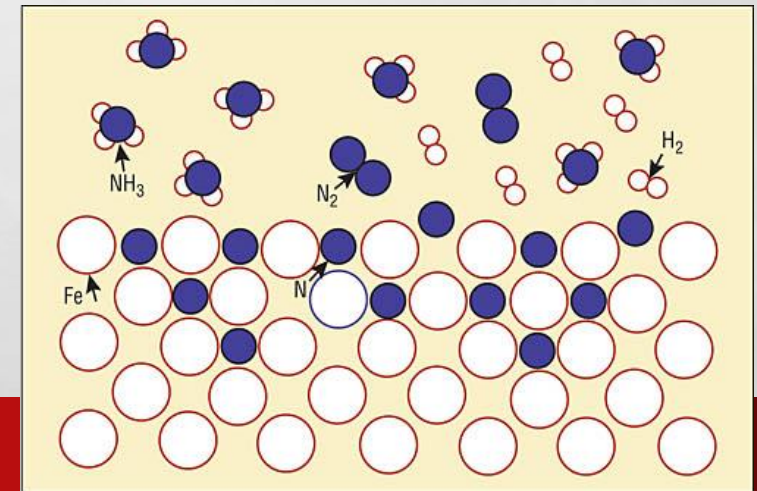
A furnace atmosphere during gas carburizing

Types of heat treatment of steel

- ❑ **Nitriding** - nitrides are formed on a metal surface in a furnace with ammonia gas circulating at 500 °C over a long period of time (100 hours). It is used for finished components.

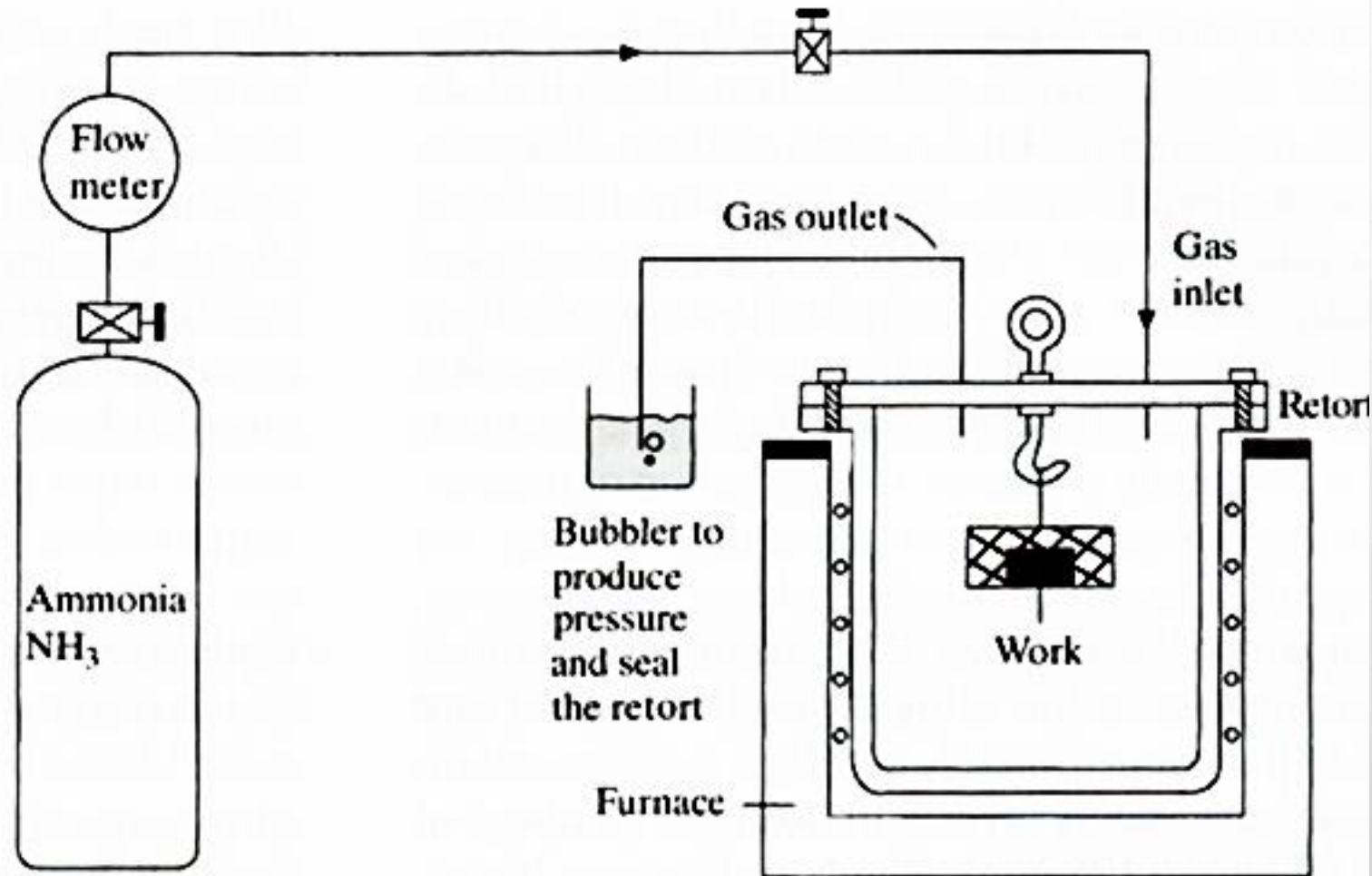
Nitriding heats the steel part to 480–620 °C in an atmosphere of ammonia gas and dissociated ammonia. The time the part spends in this environment dictates the depth of the case.

The advantage of this process is that it **causes little distortion**, so the part can be case-hardened after being quenched, tempered and machined. **No quenching** is done after nitriding and the **hardness** is achieved by the formation of nitrides.



Dissociation of ammonia and nitrogen pickup in steel during gas nitriding

Types of heat treatment of steel

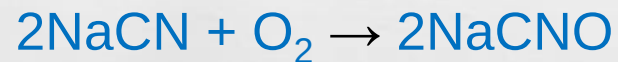


Gas nitriding system: Reduction process: $2\text{NH}_3 \rightarrow 2\text{N} + 3\text{H}_2$

Types of heat treatment of steel

Cyaniding

- ❑ **Cyaniding** is a case-hardening process that is fast and efficient; it is mainly used on low-carbon steels. The part is heated to 871-954 °C in a bath of **sodium cyanide** and then is quenched and rinsed, in water or oil, to remove any residual cyanide.



C and **N₂** diffuse into steel and give thin wear resistant layer of carbonitride phase

- ❑ This process produces a thin, hard shell (between 0.25 - 0.75 mm, 0.01 and 0.03 inches) that is harder than the one produced by carburizing, and can be completed in 20 to 30 minutes compared to several hours so the parts have less opportunity to become distorted. It is typically used on small parts such as **bolts, nuts, screws** and small **gears**.



Types of heat treatment of steel

Carbonitriding

- ❑ **Carbonitriding** is similar to cyaniding except a gaseous atmosphere of **ammonia and hydrocarbons** is used instead of **sodium cyanide**. If the part is to be quenched, it is heated to 775–885 °C; if not, then the part is heated to 649–788 °C.

Ferritic nitrocarburizing

- ❑ **Ferritic nitrocarburizing** diffuses mostly nitrogen and some carbon into the case of a workpiece below the critical temperature, approximately 650 °C. Under the critical temperature the workpiece's microstructure does not convert to an **austenitic** phase, but stays in the **ferritic** phase, which is why it is called **ferritic nitrocarburization**.

Types of heat treatment of steel

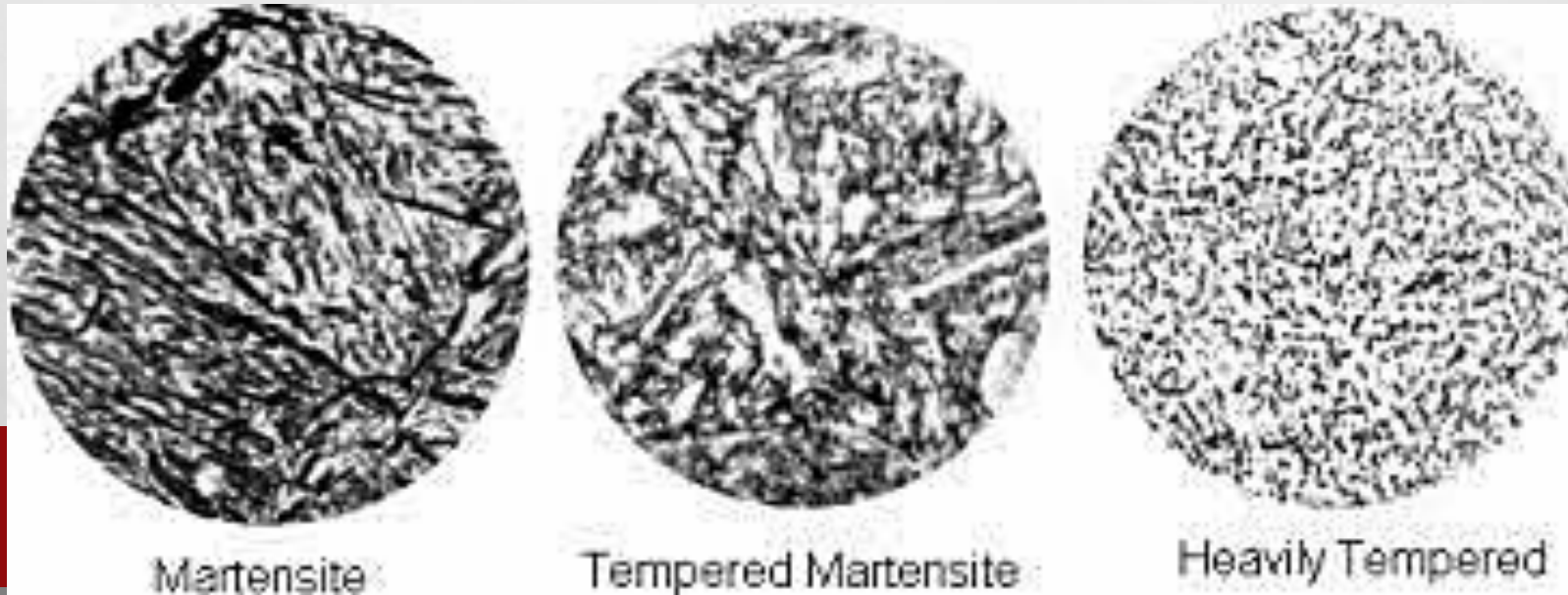
2. Tempering

- ❑ A process of heat treating, which is used to increase the **toughness** of iron-based alloys. Tempering is usually performed after **hardening**, to reduce some of the excess hardness, and is done by heating the metal to some temperature below the critical point for a certain period of time, then allowing it to cool in still air.
- ❑ Higher temperatures will result into high ductility, but low strength and hardness.
- ❑ Low tempering temperatures will produce low ductility, but high strength and hardness.

Types of heat treatment of steel

2. Tempering

Photomicrograph of martensite, a very hard microstructure formed when steel is quenched. Tempering reduces the hardness in the martensite by transforming it into various forms of tempered martensite.

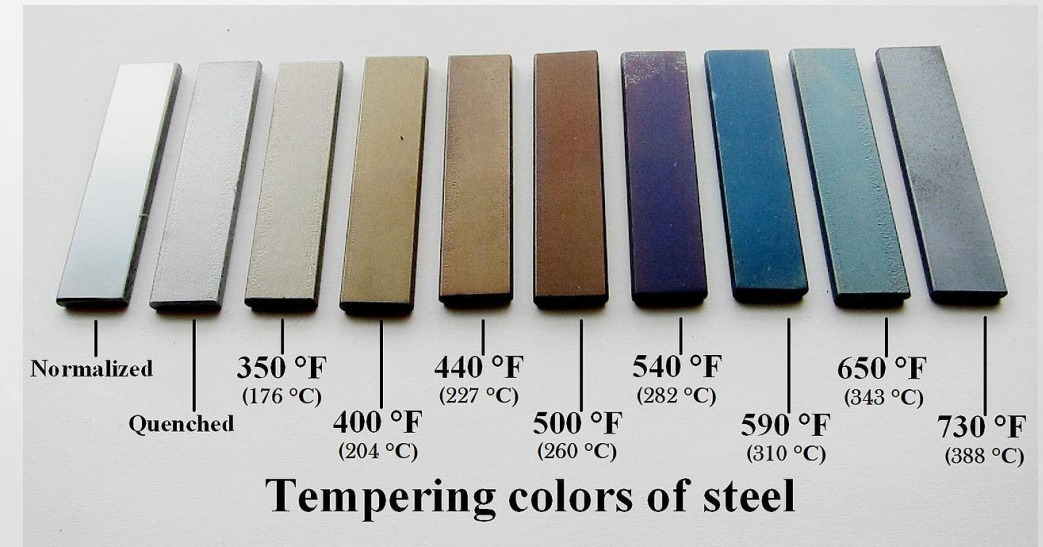


Types of heat treatment of steel

2. Tempering colors

The various colors, their corresponding temperatures, and some of their uses are:

- Faint-yellow – 176 °C (349 °F) – engravers, razors, scrapers
- Light-straw – 205 °C (401 °F) – rock drills, reamers, metal-cutting saws
- Dark-straw – 226 °C (439 °F) – scribes, planer blades
- Brown – 260 °C (500 °F) – taps, dies, drill bits, hammers, cold chisels
- Purple – 282 °C (540 °F) – surgical tools, punches, stone carving tools
- Dark blue – 310 °C (590 °F) – screwdrivers, wrenches
- Light blue – 337 °C (639 °F) – springs, wood-cutting saws
- Grey-blue – 371 °C (700 °F) and higher – structural steel



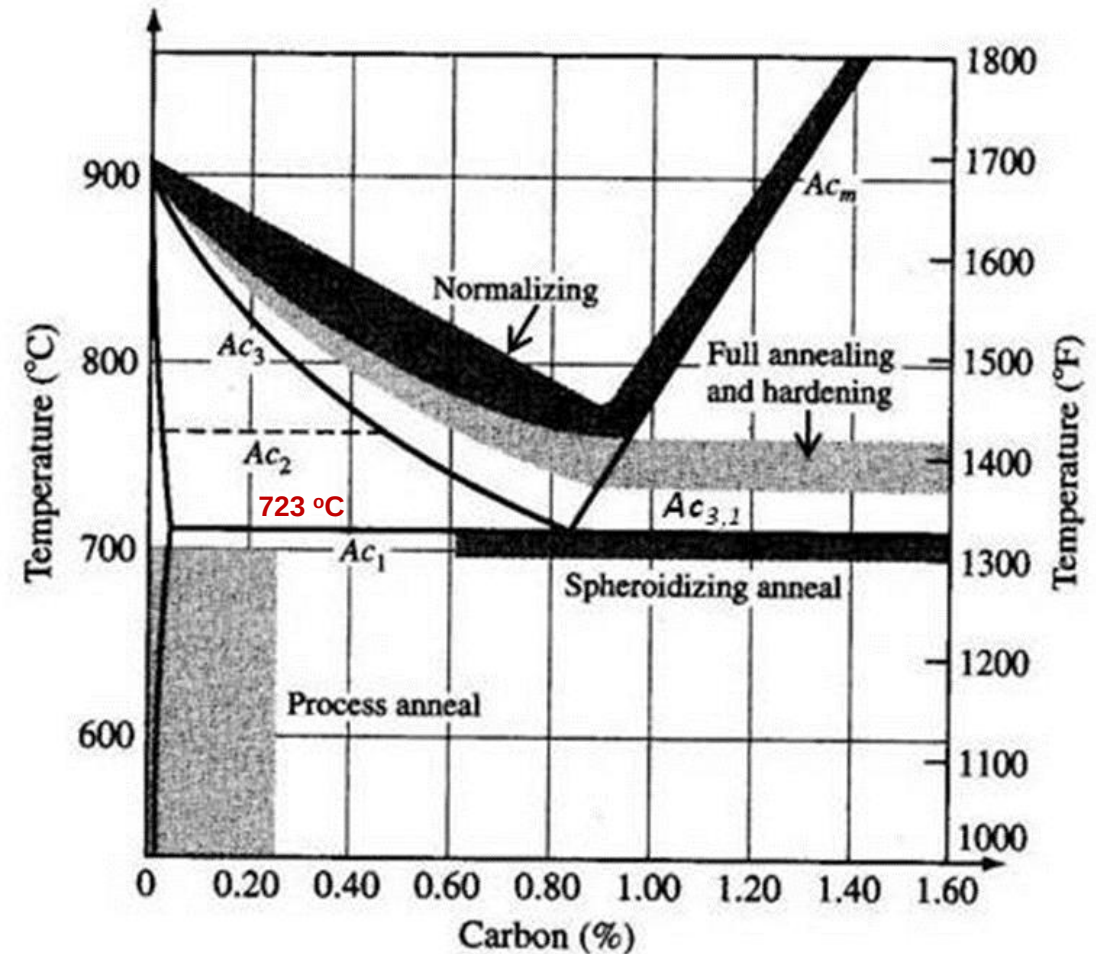
Ref. [https://en.wikipedia.org/wiki/Tempering_\(metallurgy\)](https://en.wikipedia.org/wiki/Tempering_(metallurgy))

Types of heat treatment of steel

3. Annealing

What is Annealing?

Annealing is a process by which the properties of steel are enhanced to meet machinability requirements. Annealing is a process of heating the steel slightly above the **critical temperature** of steel (723 °C) and allowing it to cool down very slowly.



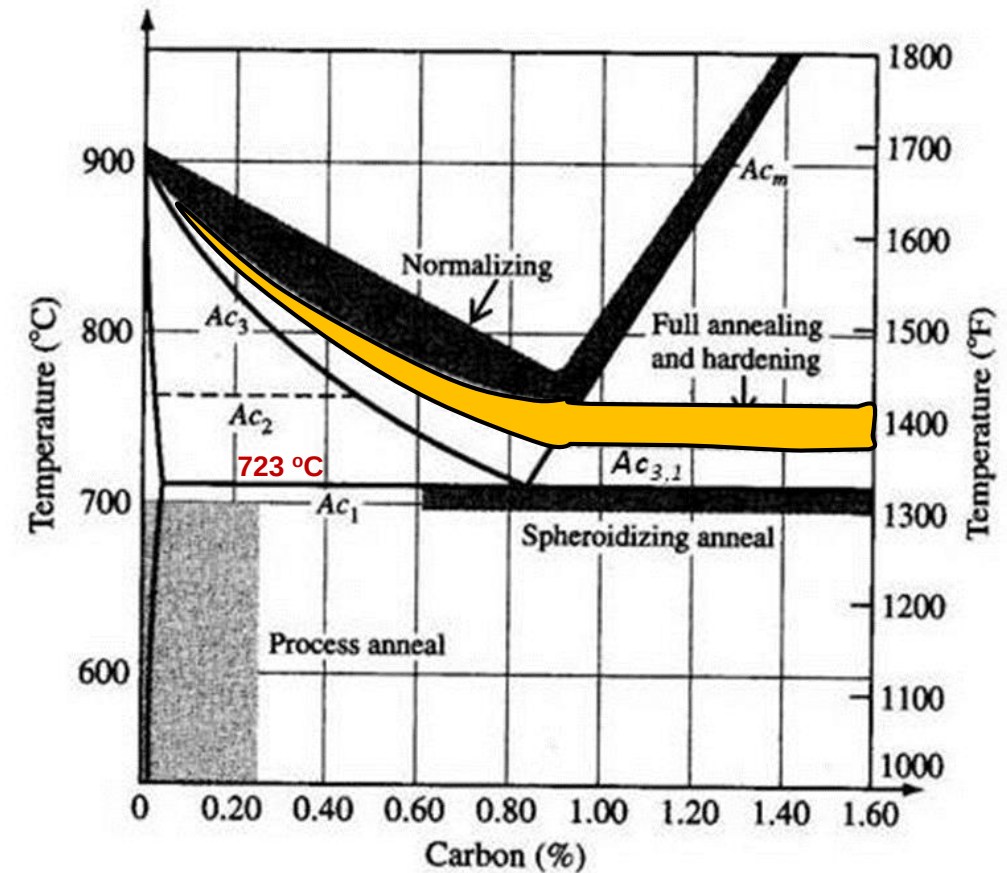
Temperature Range for Heat Treatment of Carbon Steels

Types of heat treatment of steel

Types of Annealing

There are various types of annealing:

1. **Full Annealing** - The process involves heating the steel to 30 to 50 °C above the **critical temperature** of steel and maintaining the temperature for a specified period of time, then allowing the material to slowly cool down inside the furnace itself without any forced means of cooling. Hot Worked sheets, forgings, and **castings** made from medium and high carbon **steels** need full annealing.

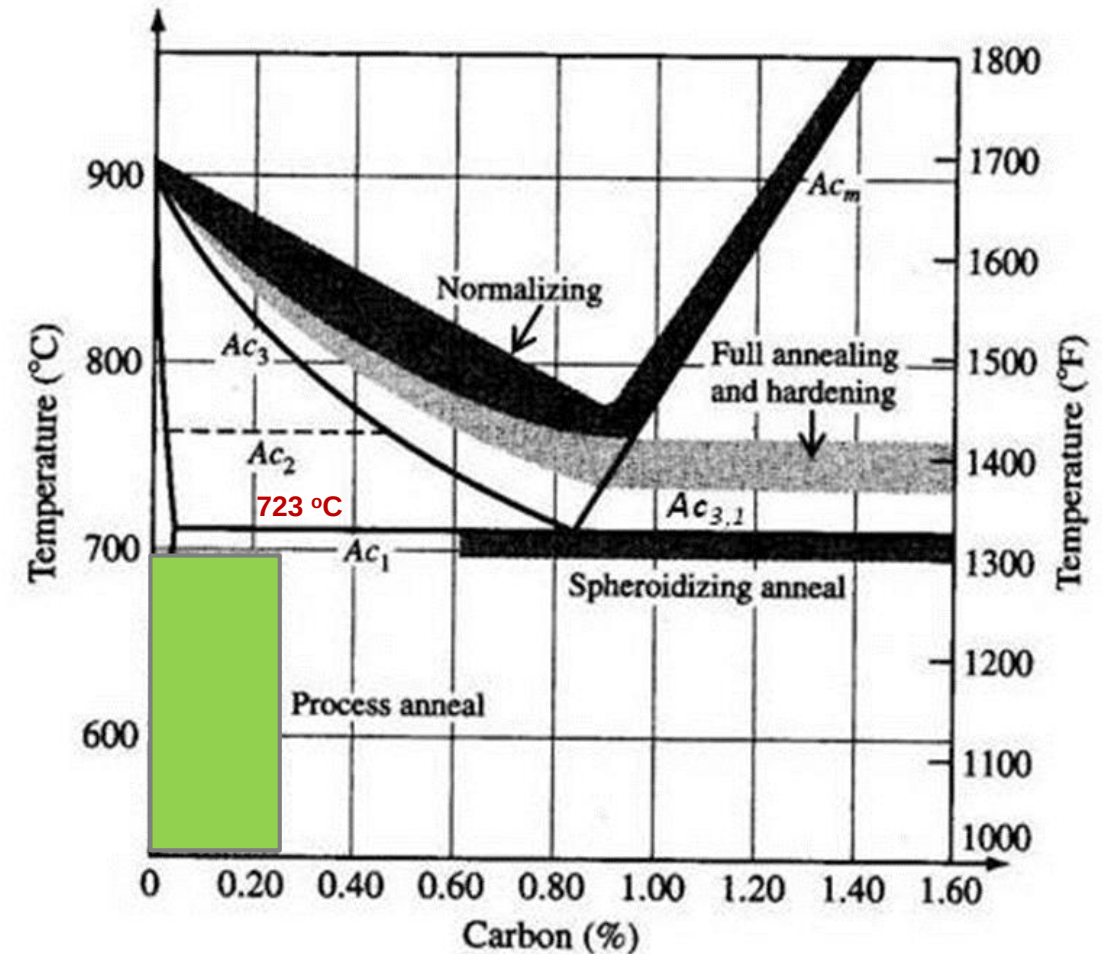


Temperature Range for Heat Treatment of Carbon Steels

Types of heat treatment of steel

Types of Annealing

2. Process Annealing - This process is mainly suited for **low carbon steel**. The material is heated up to a temperature just **below the lower critical temperature** of steel. Cold worked steel normally tends to possess increased hardness and decrease ductility making it difficult to work. Process annealing tends to improve these characteristics. This is mainly carried out on cold rolled steel like wire drawn steel, etc.



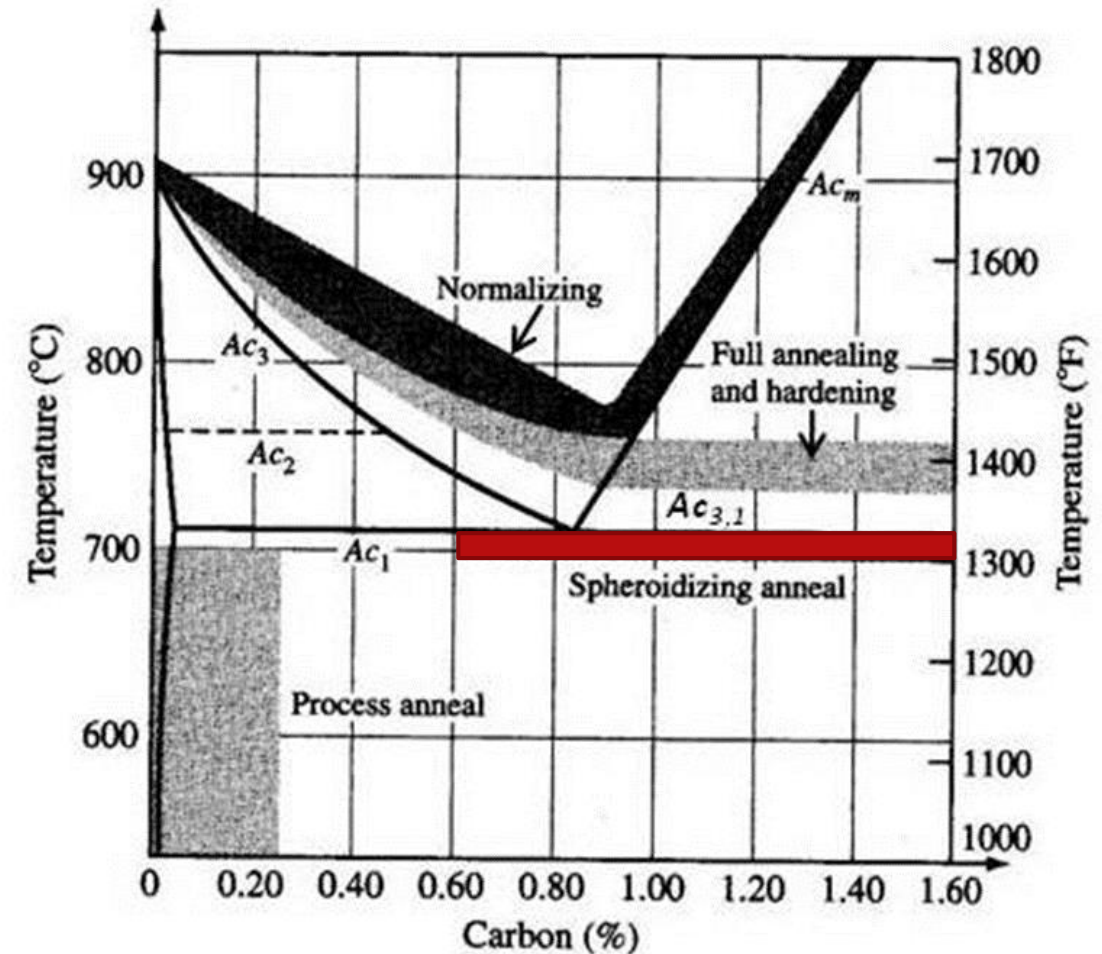
Temperature Range for Heat Treatment of Carbon Steels

Types of heat treatment of steel

Types of Annealing

3. Spherodize Annealing - This is a process for high carbon and alloy steel in order to **improve their machinability**. The process tends to improve the internal structure of the steel. **This can be done by two methods**

- A. The material is heated just **below the lower critical temperature** about 700 °C and the temperature is maintained for about 8 hours and allowed to cool down slowly.
- B. Heating and cooling the material alternatively between **temperatures just above and below the lower critical temperature**.



Temperature Range for Heat Treatment of Carbon Steels

Types of heat treatment of steel

Types of Annealing

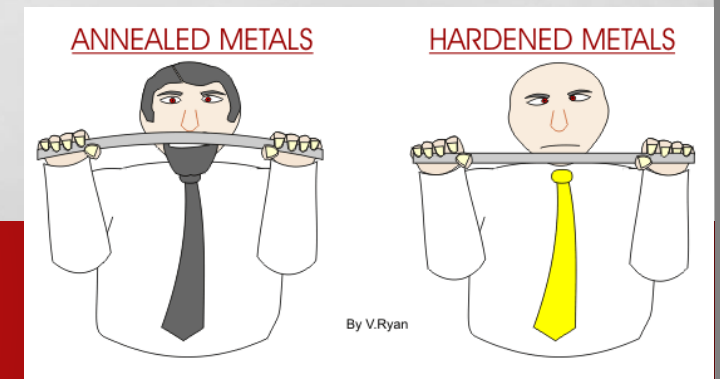
4. **Stress Relief Annealing** - Large castings or welded structures tend to possess internal stresses caused mainly during their manufacture and uneven cooling. This internal stress cause **brittleness** at isolated locations in the castings or structures, which can lead to **sudden breakage** or **failure** of the material. This process involves heating the casting or structure to about 650 °C. The temperature is maintained constantly for a few hours and allowed to cool down slowly

Types of heat treatment of steel

Advantages of Annealing

The following are some of the advantages of annealing.

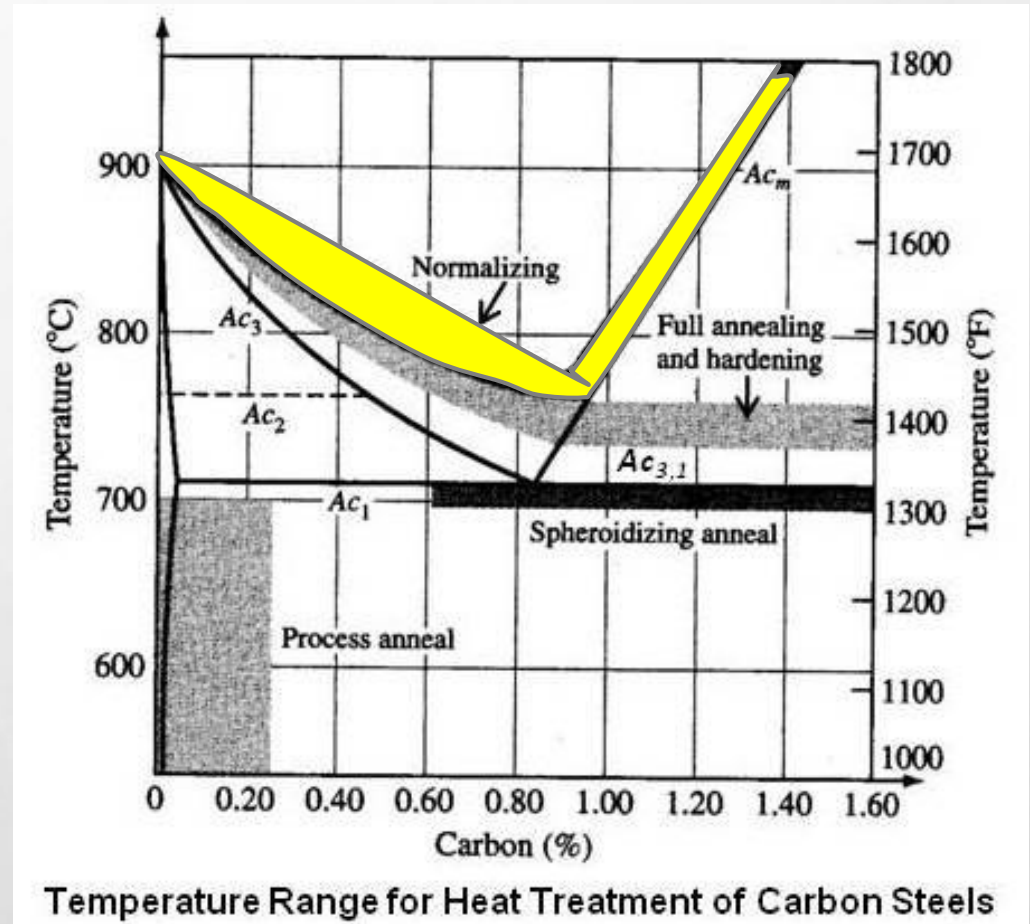
- It softens the steel.
- It enhances and improves the machinability of steel.
- It increases the ductility of steel
- It enhances the toughness of steel
- It improves the homogeneity in steel
- The grain size of the steel is refined a lot by annealing
- It prepares the steel for further [heat treatment](#).



Types of heat treatment of steel

4. Normalizing

The soaking temperature is 30-50 °C above A_3 or A_{cm} in austenite field range. The temperature depends on carbon content. After soaking the alloy is cooled in still air. This cooling rate and applied temperature produces small grain size. The small grain structure improve both toughness and strength **(especially yield strength)**.



Types of heat treatment of steel

Normalizing is normally done to achieve any one of the following purposes

- ☐ To modify and/or refine the grain structure and to eliminate coarse grained structures obtained in previous working operations such as rolling and forging etc.
- ☐ To modify and improve cast dendritic structures and reduce segregation by homogenization of the microstructure.
- ☐ To produce a homogeneous microstructure and to obtain desired microstructure and mechanical properties.
- ☐ To improve machinability of low carbon steels.
- ☐ To improve ductility and toughness.
- ☐ To provide a more consistent response when hardening or case hardening.
- ☐ To remove macrostructure created by irregular forming or by welding.